PRIVACY-PRESERVING SHOPPING INTELLIGENCE

Cloud vs federated vs on-device: what we compare, and what we know so far

Short version — supervisor meeting, 30 September 2026

Cloud Federated On-device

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THE QUESTION

Three deployments, compared on the same users

	CLOUD	FEDERATED	ON-DEVICE
What leaves the device	Raw browsing events	Model updates only	Nothing (after one download)
Model	Large (83.9M parameters)	Device-size (20.7M)	Device-size (20.7M)
Privacy	None — server sees everything	Optional differential privacy	Data never leaves
Main risk	Privacy	Quality and bandwidth	Too little data per user

The number we report: retention = method ÷ cloud, on the same users and the same test week, with a confidence interval.

Source: Research proposal §1; study description; architecture lock (parameters)

WHAT WE COMPARE

Four comparisons answer the question

REFERENCE

Cloud model

- Matched: same 131k users
- Full pool: 1.35M users (what Edge would run)

FLOOR

"Last viewed item" rule — no training, nothing uploaded

1 **Federated vs cloud**

How much quality is lost when events stay on the device?

2 **Federated + DP vs federated**

What does a formal privacy guarantee cost? (ε = 8, ε = 1)

3 **On-device vs cloud**

How much survives with no upload at all (local model, local fine-tuning)?

4 **Pretrained branch vs full-pool cloud**

Realistic Edge: pretrain on opted-in users → federated on private users.

Every method is also compared with the floor: gain over "last viewed item", with a confidence interval.

Source: Pre-registered contrast list (as revised after the independent review)

COMPARISON BOARD

Every method: what it is compared with, and where it stands

Level	Method	Compared with	Status
Cloud	Large cloud model	reference	0.1612 (contest)
Cloud	Full-pool cloud (1.35M users)	reference 2	training ends ≈ 30 Sep
Cloud	Small cloud model (device size)	large cloud	sealed
Federated	Federated (tuned), FedProx, personal vector, 8-bit uploads	cloud; floor	tuning stage 2 of 4
Federated + DP	DP at ε = 8 and ε = 1	federated	after tuning
Pretrained branch	Pretrained → + federated → + frozen tables	full-pool cloud	pretraining running
On-device	Local-only model; cloud / federated / pretrained model fine-tuned on device	cloud; its own base	device runs started
Baselines	Last viewed item 0.0711 · session kNN 0.1114 · popularity 0.0194	floor	done

All sealed results are read together: validation ≈ 2–3 Oct, single test look ≈ 4–5 Oct (estimates).

Source: Contest result; baselines (decision log 28 Sep); run status (decision log 29 Sep); master plan v5

DISCUSSION / DECISION 1

Which comparison should lead the Edge answer?

1

Which of the four comparisons should be the headline for Edge?

Plain federated vs cloud, or the realistic pretrained branch vs the full-pool cloud (the independent review recommends the latter).

2

Against which cloud: the matched 131k-user model or the full-pool 1.35M-user model?

We report both; the question is which one leads.

3

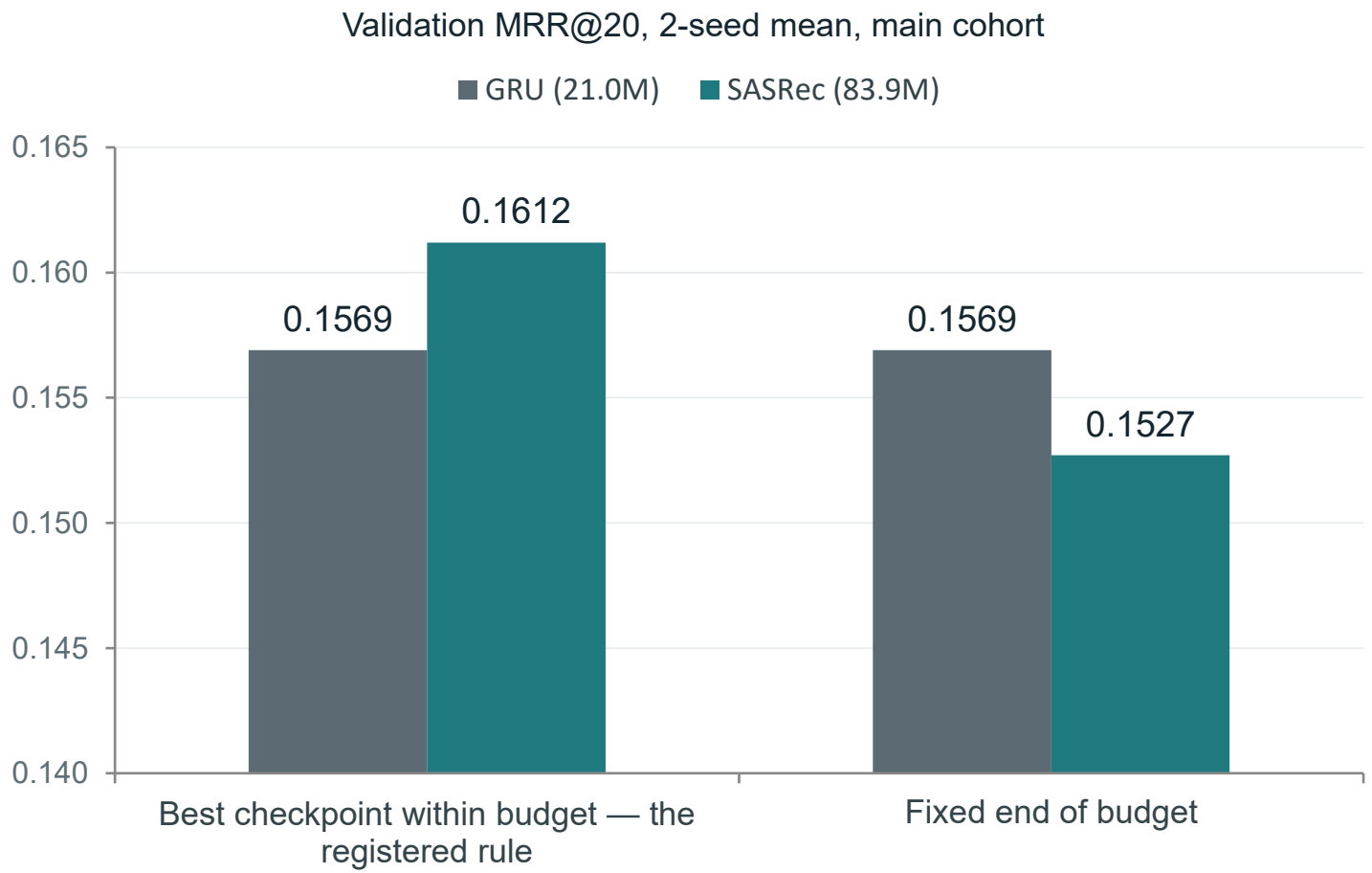
Should “last viewed item” be the floor every method must beat?

It needs no training and sends nothing.

Context: four comparisons and the floor, previous slides

COMPARISON 1 — MODEL FAMILY

SASRec vs GRU: the winner depends on the checkpoint rule



SASRec +0.0043

under the registered rule, CI [+0.0033, +0.0052] → selected

GRU +0.0042

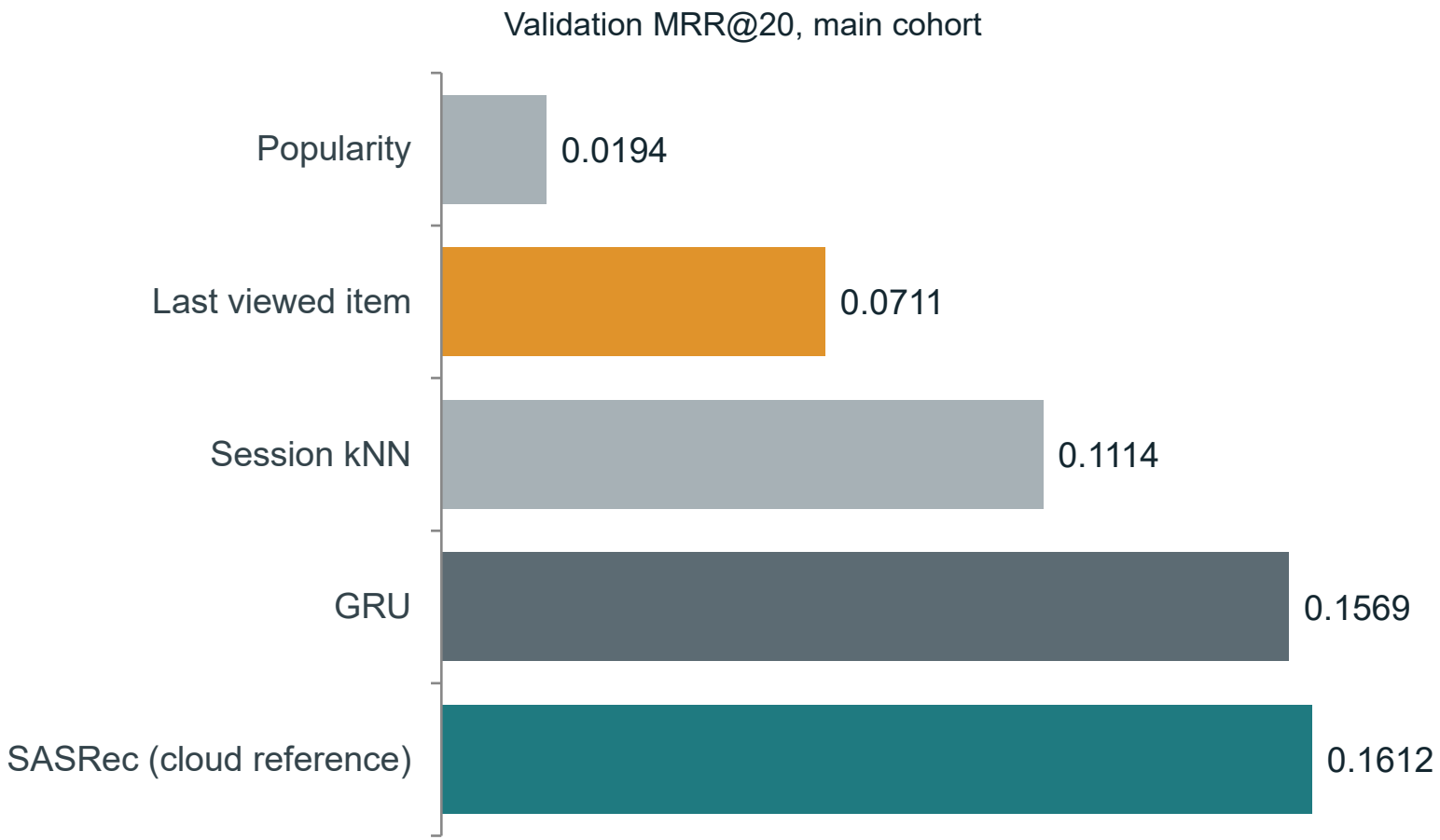
at the fixed end, CI [+0.0033, +0.0052] → disclosed

Both rules were fixed before any result. The thesis says "selected under the pre-registered rule", not "SASRec is better".

Source: Contest result file; contest diagnostics (decision log, 28 Sep)

COMPARISON 2 — MODELS VS SIMPLE RULES

What a trained model adds over simple rules



44%

of the cloud reached by “last viewed item” — for free

69%

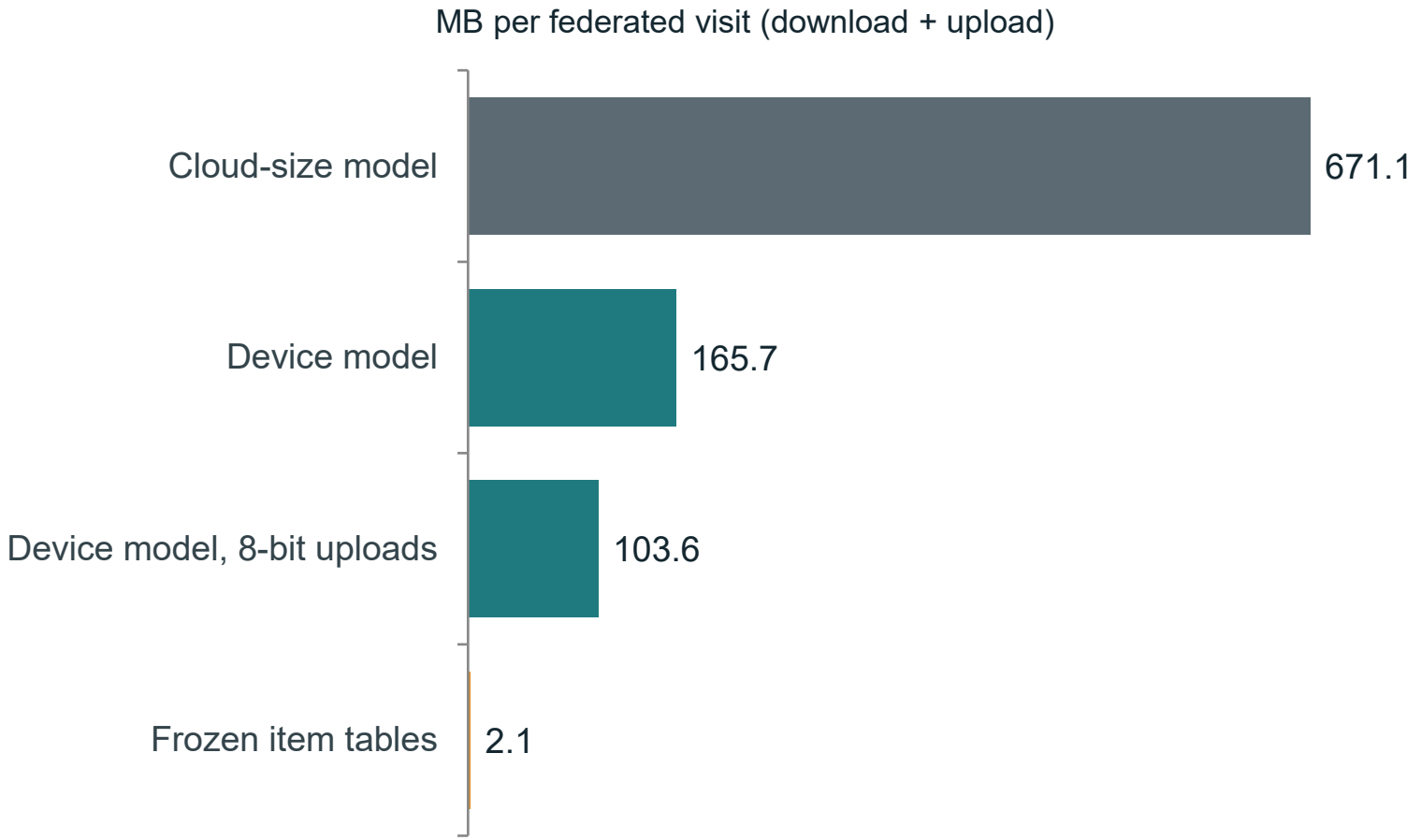
of the cloud reached by session kNN

So the bar for any private method is high: it must beat 0.0711 to be worth shipping.

Source: Decision log (post-contest reported read of baselines and contest runs, 28 Sep)

COMPARISON 3 — COST ON THE DEVICE

Cloud-size vs device-size model: megabytes per visit



170 MB

device cap per visit — only the device-size model fits

- Parameters: 83.9M cloud vs 20.7M device.
- Frozen tables: ≈ 1 MB each way plus one 81.78 MB download (to be re-measured).
- Quality of each option: sealed until the first read.

Source: Architecture lock (bytes per visit); realism audit B12; review synthesis §3.2

COMPARISON 4 — FEDERATED VS CLOUD (EARLY WARNING)

Untuned federated learning was far below the cloud

≈ 25%

Federated (best stage so far) vs a matched central model

calibration users · 0.0288 vs ≈ 0.11–0.12 (central is an estimate)

≈ 44%

“Last viewed item” vs the cloud model

main validation users · 0.0711 vs 0.1612

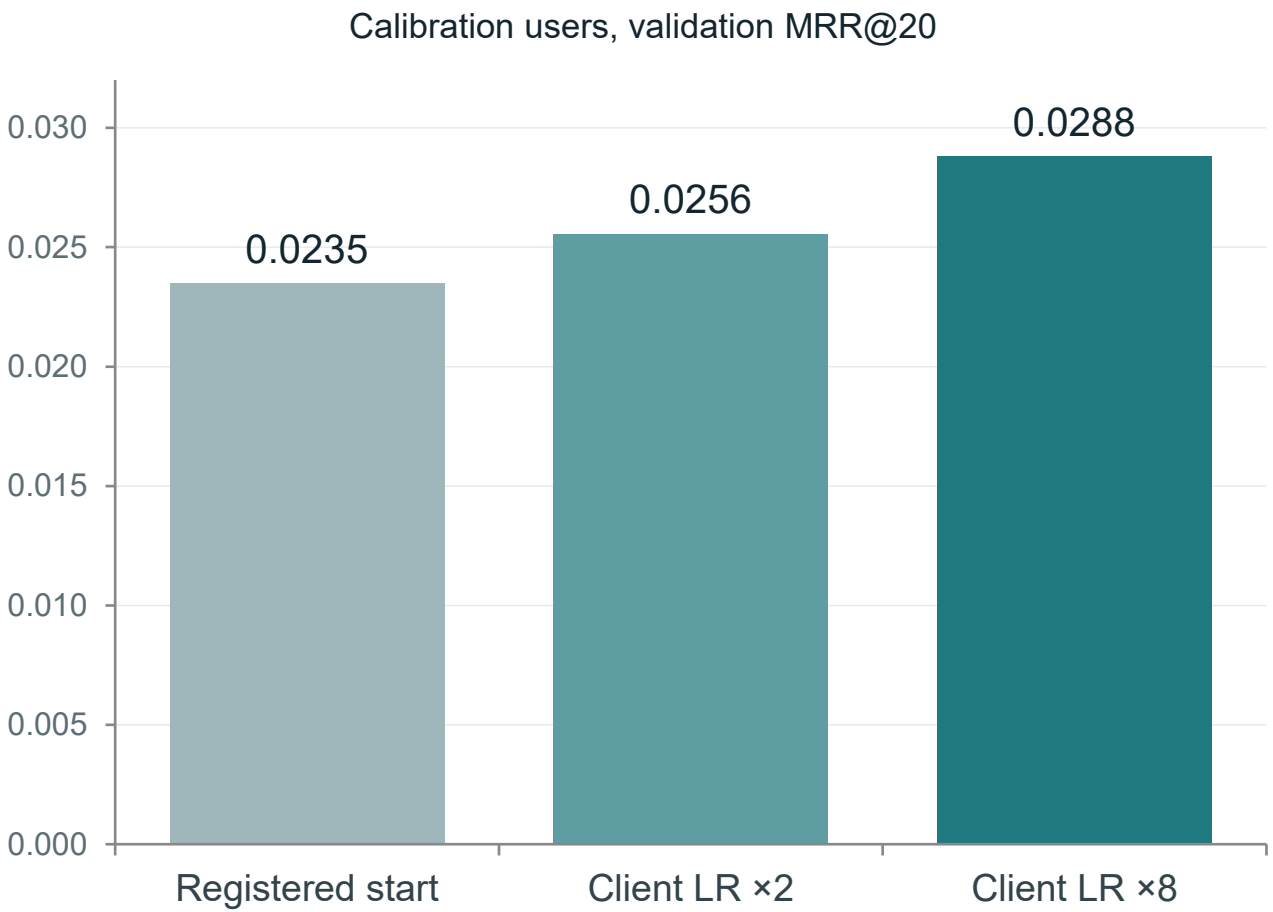
Why, and what we did

The federated recipe was the cause, not a bug: no server optimizer, a client optimizer restarted every visit, and too small a learning rate. We register a tuning ladder on calibration users only. Different users and budgets, so this is a warning, not a result.

Source: Improvement-ladder register; review synthesis §1; baselines and contest (decision log 28 Sep)

COMPARISON 5 — FEDERATED BEFORE VS AFTER TUNING

Each tuning stage vs the one before: +22.7% so far



Stages

Client learning rate	+8.9%, then +12.7%	done
Server optimizer	FedAdam / momentum	running
LR shape × client optimizer		next
Local passes and batch		next
Start from a pretrained model	pretraining	running
Freeze the item tables		next

A stage is kept only if it gains ≥ 0.002. Main-cohort values of each stage are sealed until the final read.

Source: Improvement-ladder register; ladder stage registrations; decision log 29 Sep

COMPARISON 6 — THE COST OF PRIVACY

No DP vs ε = 8 vs ε = 1, all on the same federated recipe

NO DP

ε = ∞

Updates can reveal which products a user viewed

quality: sealed

MODERATE

ε = 8

Noise multiplier 0.5642

quality: sealed

STRONG

ε = 1

Noise multiplier 1.3094

quality: sealed

more privacy →

- Guarantee: one user's whole history barely changes the released model (client-level, $\delta = 5 \times 10^{-6}$).
- Adversary: an honest-but-curious browser vendor. Secure aggregation is not simulated — stated as a limitation.

Source: Privacy statement and threat model (registered); independent check of the DP noise multipliers

DISCUSSION / DECISION 2

How far should each comparison go?

1

Is ε = 8 and ε = 1 enough, or do you want a privacy curve?

Adding ε = 4 costs one more registered run (about half a day, estimate).

2

How much tuning effort is fair for federated learning?

Our ladder stops when a stage gains < 0.002. Stop earlier and spend the time on the pretrained branch?

3

If tuned federated is still below “last viewed item”, how do we present it?

Our plan: report it plainly as a finding about this recipe and budget.

Context: comparisons 4–6

PROPOSAL VS NOW

What changed, in one view

	Proposal	Now
Dataset	Criteo, MovieLens, Amazon, Taobao, RetailRocket	REES46 shopping events (per-user, timestamped)
Task	Next category, purchase, offers	Next product viewed, over all 158,486 products
Metric	NDCG@10, HR@10, AUC	MRR@20 per user (+ NDCG@10, HR@10)
Federated	FedAvg, FedProx, FedBN, personalised	FedAvg, FedProx, personalised, 8-bit; FedBN n/a (layer norm)
Privacy	“Various levels” (undefined)	DP at ε = 8 and ε = 1; privacy matrix

Kept: cloud upper bound, federated and on-device variants, cold start, personas, communication cost, browser check.

Source: Research proposal §§3–4; study description; realism audit §F

WHEN THE COMPARISONS ARE READ

Four milestones (dates are estimates)

≈ 30 Sep

Tuning and pretraining finish

Full-pool cloud, pretraining, federated tuning stages

running

≈ 30 Sep – 2 Oct

Main federated and DP runs

Each run ≈ 10.7 h (measured rate)

next

≈ 2–3 Oct

One validation read

All methods read together, then everything is locked

sealed

≈ 4–5 Oct

Single test look → final tables ≈ 6–7 Oct

The test week is opened once

sealed

The plan said 1–2 Oct for the first read; it slipped about a day because tuning screens run slower when sharing GPUs.

Source: Master plan v5 (cost model, test-look duration); ladder schedule; decision log 29 Sep

DISCUSSION / DECISION 3

Priorities and next steps

1

If time is short, which comparison do we protect first?

Never cut: federated ×2 seeds, local-only ×2, cloud fine-tune ×2, INT8, the lock, the test look. First to cut: extra panel seeds, the 8-bit second seed, ε = 1.

2

Is a server-CPU benchmark plus one browser run on a laptop enough device evidence?

Or should we add a second machine?

3

Can you pre-approve the Queen's AI-use disclosure?

The independent review lists it as required front matter.

Context: cut order is pre-registered in the master plan